

Exercise 1.

The probability that a cell in a wireless system is overloaded is $1/3$. Given that it is overloaded, the probability of a blocked call is 0.3. Given that it is not overloaded, the probability of a blocked call is 0.1. Find the conditional probability that the system is overloaded given that your call is blocked.

Exercise 2.

The binary channel shown in Figure operates as follows. Given that a 0 is transmitted, the conditional probability that a 1 is received is ϵ . Given that a 1 is transmitted, the conditional probability that a 0 is received is δ . Assume that the probability of transmitting a 0 is the same as the probability of transmitting a 1. Given that a 1 is received, find the conditional probability that a 1 was transmitted. Hint: Use the notation $T_i := \{i \text{ is transmitted}\}, i = 0, 1$, and $R_j := \{j \text{ is received}\}, j = 0, 1$.

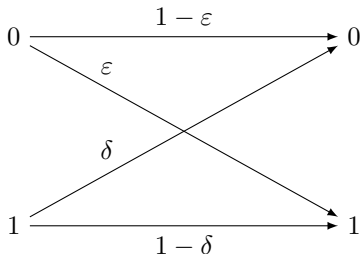


Figure: Binary channel with crossover probabilities ϵ and δ .

Exercise 3.

Fred is answering a multiple-choice problem on an exam, and has to choose one of n options (exactly one of which is correct). Let K be the event that he knows the answer, and R be the event that he gets the problem right (either through knowledge or through luck). Suppose that if he knows the right answer he will definitely get the problem right, but if he does not know then he will guess completely randomly. Let $P(K) = p$.

- Ⓐ Find $P(K|R)$
- Ⓑ Show that $P(K|R) \geq p$, and explain why this makes sense intuitively. When (if ever) does $P(K|R)$ equal p ?

Exercise 4.

A hat contains 100 coins, where 99 are fair but one is double-headed (always landing Heads). A coin is chosen uniformly at random. The chosen coin is flipped 7 times, and it lands Heads all 7 times. Given this information, what is the probability that the chosen coin is double-headed?

Exercise 5.a.

Alice is trying to communicate with Bob, by sending a message (encoded in binary) across a channel.

(a) Suppose for this part that she sends only one bit (a 0 or 1), with equal probabilities. If she sends a 0, there is a 5% chance of an error occurring, resulting in Bob receiving a 1; if she sends a 1, there is a 10% chance of an error occurring, resulting in Bob receiving a 0. Given that Bob receives a 1, what is the probability that Alice actually sent a 1?

Exercise 5.b

(b) To reduce the chance of miscommunication, Alice and Bob decide to use a repetition code. Again Alice wants to convey a 0 or a 1, but this time she repeats it two more times, so that she sends 000 to convey 0 and 111 to convey 1. Bob will decode the message by going with what the majority of the bits were. Assume that the error probabilities are as in (a), with error events for different bits independent of each other. Given that Bob receives 110, what is the probability that Alice intended to convey a 1?